DDPM and DDIM

- DEs:
  - Forward process:

    $dx = f(x,t)dt + g(t)dw$ where:
    - dx: In very short time, how does the image change.
    - $f(x,t)dt$: Drift term. A term that pulls the pixels to the origin $\mu = 0$.
    - $g(t)dw$: Diffusion term. $g(t)$ is the term that controls the strength of noise, w is the Brownian motion, and dw means the gaussian noise injected in a very short time dt.

    > Original image is $x(0)$. With the effect of the drift term and the gaussian noise, the image becomes gaussian noise at $t = 1$.

  - Anderson's theorem and the reverse process.
    - For every SDE, there is a reverse SDE.
    - $dx = \left[f(x,t) - g(t)^2 \nabla_x \log p_t(x)\right] dt + g(t) d\bar{w}$, *can be conducted using Brian D. O. Anderson's work*.
      - $d\bar{w}$: Reversed Brownian motion.
      - $-g(t)^2 \nabla_x \log p_t(x)$: **Score Function**. $p_t(x)$ is the probability of ALL images. $\nabla_x \log p_t(x)$ is its gradient. **This is what we predict, i.e.,** $\nabla_x \log p_t(x) \approx -\frac{\epsilon_\theta(x_t,t)}{\sigma_t}$.

      > During this process, there is not only brownian motion but also a Score Function that pushes the data to the real point.

  - DDIM reverse process: Directly remove the reverse Brownian motion term:

    $dx = \left[f(x,t) - \frac{1}{2} g(t)^2 \nabla_x \log p_t(x)\right] dt$
    - This is an ODE.
    - If the starting point, the noise $x(1)$ is given (i.e., use a fixed seed), then the generated image $x(0)$ is deterministic.
    - Fast: No need to solve an SDE, so we can use matured ODE solvers.
- Actual Implementation:
  - General form:

    $$x_{t-1} = \underbrace{\sqrt{\bar{\alpha}_{t-1}} \left(\frac{x_t - \sqrt{1-\bar{\alpha}_t}\epsilon_\theta(x_t,t)}{\sqrt{\bar{\alpha}_t}} \right)}_{\text{Term 1: 预测的清晰原图 } x_0} + \underbrace{\sqrt{1 - \bar{\alpha}_{t-1} - \sigma_t^2} \cdot \epsilon_\theta(x_t,t)}_{\text{Term 2: 指向当前状态的梯度方向}} + \underbrace{\sigma_t \epsilon}_{\text{Term 3: 随机噪声}}$$
    - Term 1: Target based on the current state x_t and the current predicted noise (x_0).
    - Term 2: The term that "pulls" the image "back" to the $t-1$ state.
    - Term 3: Random noise for avoiding blurry.
  - DDPM: Let: $\sigma_t = \sqrt{\frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}} \sqrt{1 - \frac{\bar{\alpha}_t}{\bar{\alpha}_{t-1}}}$ (a non-zero value come from Markov)
  - DDIM: Let: $\sigma_t = 0$

- The loss function, for both DDPM and DDIM, is the same: $L = \mathbb{E}_{t,x_0,\epsilon\sim\mathcal{N}(0,I)}\left[\|\epsilon - \epsilon_\theta(x_t,t)\|^2\right]$.
- Why can Term 3 help the result to be clear and sharp in **DDPM**?
  - Because of MSE, the network tends to output **expected** noise, which will make the result blurry.
  - With the random small noise, it will "push" the system away from the **average,** (Based on Langevin Dynamics, it injects a variance to the system), so it will reach a specific, clear image.
- Why can DDIM produce clear images without Term 3?
  - This is based on the following 2 properties of ODEs:

  1. They have continuous, smooth vector fields;
  2. If i. is fulfilled, then it is a bijective mapping (Picard-Lindelöf Theorem).

  Meaning that if a noisy image ($t = T$) is provided, then it will be converted to a specific point on the actual distribution. It is impossible for multiple different initial noise values x_T to eventually converge to a single fuzzy point (unless that point is a center of attraction; in the score field constructed by Diffusion, the data manifold containing the actual high-resolution images is the center of attraction).
- How to connect the implementation and the DE representation? (Why do we let $\sigma_t = 0$ correspond to the ODE/DDIM and $\sigma_t = \ldots$ correspond to SDE/DDPM?)

  $x_{t-1} - x_t \approx \underbrace{\Phi(x_t, t, \epsilon_\theta)\cdot\Delta t}_{\text{确定性位移 (Drift)}} + \underbrace{\sigma_t\epsilon}_{\text{随机位移 (Noise)}}$ by rearrange

  $dx = \underbrace{[\ldots\text{漂移项}\ldots]dt}_{\text{确定性位移 (Drift)}} + \underbrace{g(t)d\bar{w}}_{\text{随机位移 (Noise)}}$

  If Δt is very small, then $d\bar{w} \approx \sqrt{\Delta t}\cdot\epsilon$ (Based on Euler-Maruyama method of SDE)

  So when Δt is very small, i.e., $\approx dt$, then $\begin{aligned}\sigma_t\epsilon = g(t)d\bar{w} = g(t)\sqrt{\Delta t}\cdot\epsilon \\ \sigma_t = g(t)\sqrt{\Delta t}\end{aligned}$

  In this case, if we let $\sigma_t = 0$, then $g(t)d\bar{w}$ will be canceled out. Then, the system will become an ODE (DDIM).
- ELBO and MSE

  For simplicity, we represent:

  $\text{Loss} = -\text{ELBO} = L_T + \sum_{t=2}^{T} L_{t-1} + L_0$

  We only care about $L_{t-1} = D_{KL}\left(\underbrace{q(x_{t-1}|x_t,x_0)}_{\text{真实的逆向去噪分布}}\middle\| \underbrace{p_\theta(x_{t-1}|x_t)}_{\text{神经网络预测的去噪分布}}\right)$ as L_T is a constant, and L_0 will be commonly simplified by replacing it with a constant.

  In DDPM, no matter the actual reverse distribution q or predicted distribution p_θ, they are both gaussian distributions with a fixed variance (depends on β_t)

  With this in mind, we have:

  $L_{t-1} \propto \|\tilde{\mu}_t - \mu_\theta\|^2$, where

  $\tilde{\mu}_t$ is the actual distribution's mean, and

  μ_θ is the predicted distribution's mean

Based on Bayes theorem, we have:

$\tilde{\mu}_t(x_t, x_0) = \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1-\bar{\alpha}_t}x_0 + \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}x_t$

However, at this moment, the model has no idea about x_0, so from the forward process:

$x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t}\epsilon$

We can represent x_0:

$x_0 = \frac{x_t - \sqrt{1-\bar{\alpha}_t}\epsilon}{\sqrt{\bar{\alpha}_t}}$

Bring this back and rearrange it:

$\tilde{\mu}_t = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon\right)$

If the actual mean is like this, we want to let the network align with it, so without changing the form, replace ϵ with what it predicted:

$\mu_\theta = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon_\theta(x_t, t)\right)$

Then we have:

$L_{t-1} \propto \frac{\beta_t^2}{2\sigma_t^2\alpha_t(1-\bar{\alpha}_t)}\|\epsilon - \epsilon_\theta(x_t, t)\|^2$

Based on the experiment, the researchers found that letting the complicated coefficient be 0 will also work (even better), so the final form is:

$L_{simple} = \mathbb{E}_{t,x_0,\epsilon}\left[\|\epsilon - \epsilon_\theta(x_t, t)\|^2\right]$